FIT9132 Applied Week 4

A4-1 The Relational Model

A41.1 Relational Model Theory

1) **Relation** : A set of attributes consists of a head and a body,
   The heading is the schema, and the body is the sets of tuples.
2) **Attribute** : A characteristic of an objective that we want to record
3) **Domain** : A set of atomic values from which attribute's value are drawn. Domain consists of a name, data type and data format. For example: cust_age: NUMBER(3)
4) **Tuple** : A set of related values that describe an instance of the relation (It is a row/ observation)
5) **Degree and Cardinality of a relation** :
   a) **Degree** : Number of attributes
   b) **Cardinality** : Number of Tuples
6) **Primary Key, Super Key and Foreign Key**
   a) **Super Key** : A set of attributes that uniquely define a tuple.
   b) **Primary Key** : Unique identifier (An attribute or a MINIMAL set of attributes that uniquely identifies each tuple).
   c) **Foreign Key** : An attributes in a relation that exists in the same or another relation as Primary Key.

A4-1.2 Choosing the Primary Key

1)

2) Identify the primary key and foreign key for these three relations:
   ORDER (order_id, order_date, cust_id)
   Primary Key : order_id
   Foreign Key : None

   ORDERLINE (order_id, prod_no, ol_qtyordered, ol_lineprice)
   Primary Key : order_id
   Foreign Key : order_id, prod_no

   PRODUCT (prod_no, prod_desc, prod_unitprice)
   Primary Key : prod_no
   Foreign Key : None

3) **APPOINTMENT (dentist_id, dentist_name, patient_id, patient_name, appointment_datetime, surgery_roomno)**
Identify the superkey(s), candidate key(s) and the primary key for the relation if the following business rules are applicable:

a) A dentist can only see a single patient at a particular date and time
b) A dentist treats a patient in a particular surgery room (only one patient is treated in the room at a time), and
c) A patient can see the same dentist multiple times.

Candidate Key :
dentist_id, app_datetime
patient_id, app_datetime
sur_roomno, app_datetime

Primary Key :
Dentist_id

Super Key :
dentist_id, appointment_datetime (because of rule 1)
surgery_roomno, appointment_datetime (because of rule 2)

4.2 Relational Algebra

4.2.1 Relational Algebra Exercise

HOTEL (hotel_no, hotel_name, hotel_city)

ROOM (room_no, hotel_no, room_type, room_price)

BOOKING (hotel_no, guest_no, bdate_from, bdate_to, room_no)

GUEST (guest_no, guest_name, guest_address)

$\pi \bowtie \sigma$

1) **List the number and name for all hotels**

   A1 = π hotel_no, hotel_name HOTEL

2) **List all single rooms with a price below $50**

   A2 = σ room_type = 'single' and room_price < 50 ROOM

3) **List the numbers and names of all hotels in Melbourne**

   A3 = π hotel_no, hotel_name (σ hotel_city = 'Melbourne' HOTEL)

4) List all numbers and names of hotels which have a presidential suite room

   A4a = π hotel_no (σ room_type = 'presidential suite' ROOM)
   A4b = A4a $\bowtie$ π hotel_no, hotel_name HOTEL

   A4 = π hotel_no, hotel_name HOTEL(π hotel_no (σ room_type = 'presidential suite' ROOM))

5) List the price and type of all rooms at the Grosvenor Hotel

6) List all numbers, names, and addresses of guests currently staying in deluxe room of any hotel (assume that if the guest has a tuple in the BOOKING relation, then they are currently staying in the hotel)

7) List all numbers, names, and addresses of guests currently staying at the Grosvenor Hotel (assume that if the guest has a tuple in the BOOKING relation, then they are currently staying in the hotel)